Q1 - 2024 (04 Apr Shift 1)

One of the points of intersection of the curves $y=1+3x-2x^2$ and $y=\frac{1}{x}$ is $\left(\frac{1}{2},2\right)$. Let the area of the region enclosed by these curves be $\frac{1}{24}(l\sqrt{5}+\mathrm{m})-\mathrm{n}\log_{\mathrm{e}}(1+\sqrt{5})$, where $l,\ \mathrm{m},\mathrm{n}\in\mathbf{N}$. Then $l+\mathrm{m}+\mathrm{n}$ is equal to

(1) 29

(2) 31

(3) 30

(4) 32

Q2 - 2024 (04 Apr Shift 2)

The area (in sq. units) of the region described by $\left\{(x,y): y^2\leq 2x\text{, and } y\geq 4x-1\right\}$ is

(1) $\frac{11}{32}$

(2) $\frac{8}{9}$

(3) $\frac{11}{12}$

(4) $\frac{9}{32}$

Q3 - 2024 (05 Apr Shift 1)

The area of the region enclosed by the parabolas $y=x^2-5x$ and $y=7x-x^2$ is

Q4 - 2024 (05 Apr Shift 2)

The area enclosed between the curves $y=x|x|$ and $y=x-|x|$ is :

(1) $\frac{4}{3}$

(2) 1

(3) $\frac{2}{3}$

(4) $\frac{8}{3}$

Q5 - 2024 (06 Apr Shift 1)

Let the area of the region enclosed by the curves $y=3x, 2y=27-3x$ and $y=3x-x\sqrt{x}$ be A. Then $10A$ is equal to

(1) 172

(2) 162

(3) 154

(4) 184

Q6 - 2024 (06 Apr Shift 2)

If the area of the region $\left\{(x, y): \frac{\mathrm{a}}{x^2} \leq y \leq \frac{1}{x}, 1 \leq x \leq 2, 0<\mathrm{a}<1\right\}$ is $\left(\log_{\mathrm{e}} 2\right)-\frac{1}{7}$ then the value of $7\mathrm{a}-3$ is equal to:

(1) 0

(2) 2

(3) -1

(4) 1

Q7 - 2024 (08 Apr Shift 1)

Let $f(x)$ be a positive function such that the area bounded by $y=f(x), y=0$ from $x=0$ to $x=a>0$ is $e^{-a}+4a^2+a-1$. Then the differential equation, whose general solution is $y=c_1 f(x)+c_2$, where c_1 and c_2 are arbitrary constants, is

(1) $\left(8e^x-1\right) \frac{d^2y}{dx^2}+\frac{dy}{dx}=0$

(2) $\left(8e^x-1\right) \frac{d^2y}{dx^2}-\frac{dy}{dx}=0$

(3) $\left(8e^x+1\right) \frac{d^2y}{dx^2}-\frac{dy}{dx}=0$

(4) $\left(8e^x+1\right) \frac{d^2y}{dx^2}+\frac{dy}{dx}=0$

Q8 - 2024 (08 Apr Shift 1)

Let the area of the region enclosed by the curve $y=\min\{\sin x, \cos x\}$ and the x axis between $x=-\pi$ to $x=\pi$ be A. Then A^2 is equal to ____________

Q9 - 2024 (08 Apr Shift 2)

The area of the region in the first quadrant inside the circle $x^2+y^2=8$ and outside the parabola $y^2=2x$ is equal to :

(1) $\frac{\pi}{2}-\frac{1}{3}$

(2) $\pi-\frac{1}{3}$

(3) $\frac{\pi}{2}-\frac{2}{3}$

(4) $\pi-\frac{2}{3}$

Q10 - 2024 (09 Apr Shift 1)

The parabola $y^2=4x$ divides the area of the circle $x^2+y^2=5$ in two parts. The area of the smaller part is equal to:

(1) $\frac{1}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(2) $\frac{1}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(3) $\frac{2}{3}+5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

(4) $\frac{2}{3}+\sqrt{5}\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$

Q11 - 2024 (09 Apr Shift 2)

The area (in square units) of the region enclosed by the ellipse $x^2+3y^2=18$ in the first quadrant below the line $y=x$ is

(1) $\sqrt{3}\pi-\frac{3}{4}$

(2) $\sqrt{3}\pi+1$

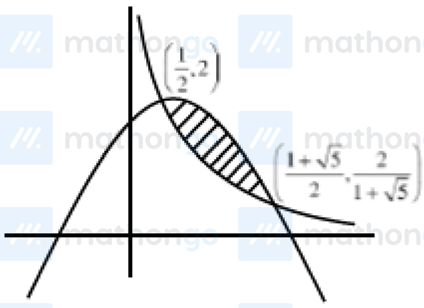
(3) $\sqrt{3}\pi$

(4) $\sqrt{3}\pi+\frac{3}{4}$

Answer Key

Q1 (3)	**Q2** (4)	**Q3** (72)	**Q4** (1)
Q5 (2)	**Q6** (3)	**Q7** (4)	**Q8** (16)
Q9 (4)	**Q10** (3)	**Q11** (3)	

Q1



$$A=\int_{\frac{1}{2}}^{\frac{1+\sqrt{5}}{2}}\left(1+3x-2x^2-\frac{1}{x}\right)\mathrm{dx}$$

$$A=\left[x+\frac{3x^2}{2}-\frac{2x^3}{3}-\ln x\right]_{\frac{1}{2}}^{1+\sqrt{5}}$$

$$A=\frac{1+\sqrt{5}}{2}+\frac{3}{2}\left(\frac{1+\sqrt{5}}{2}\right)^2-\frac{2}{3}\left(\frac{1+\sqrt{5}}{2}\right)^3-\ln\left(\frac{1+\sqrt{5}}{2}\right)$$

$$-\frac{1}{2}-\frac{3}{2}\left(\frac{1}{4}\right)+\frac{2}{3}\left(\frac{1}{8}\right)+\ln\left(\frac{1}{2}\right)$$

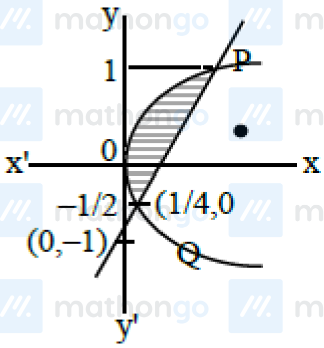
$$A=\frac{1}{2}+\frac{\sqrt{5}}{2}+\frac{3}{8}+\frac{3}{4}\sqrt{5}+\frac{15}{8}-\frac{4}{3}-\frac{2}{3}\sqrt{5}$$

$$-\frac{1}{2}-\frac{3}{8}+\frac{1}{12}-\ln(1+\sqrt{5})$$

$$=\sqrt{5}\left(\frac{1}{2}+\frac{3}{4}-\frac{2}{3}\right)+\frac{15}{8}-\frac{4}{3}+\frac{1}{12}-\ln(1+\sqrt{5})$$

$$=\frac{14}{24}\sqrt{5}+\frac{15}{24}-\ln(1+\sqrt{5})$$

Q2



$$\text{Shaded area } = \int_{-\frac{1}{2}}^{1}\left(x_{\text{Right }}-x_{\text{Left }}\right) d y$$

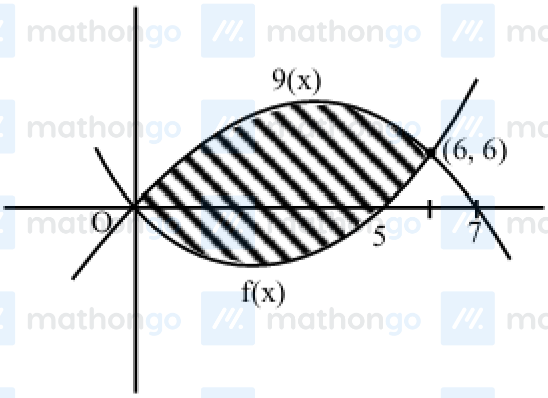
$y^2=2x$
$y=4x-1$ Solve
$y=1, y=-\frac{1}{2}$

$$\text{Shaded area } = \int_{-\frac{1}{2}}^{1}\left(\frac{y+1}{4}-\frac{y^2}{2}\right) d y$$

$$=\left(\frac{1}{4}\left(\frac{y^2}{2}+y\right)-\frac{y^3}{6}\right)_{-\frac{1}{2}}^{1}=\frac{9}{32}$$

Q3

$y=x^2-5x$ and $y=7x-x^2$



$$\int_0^6 (g(x) - f(x))dx$$

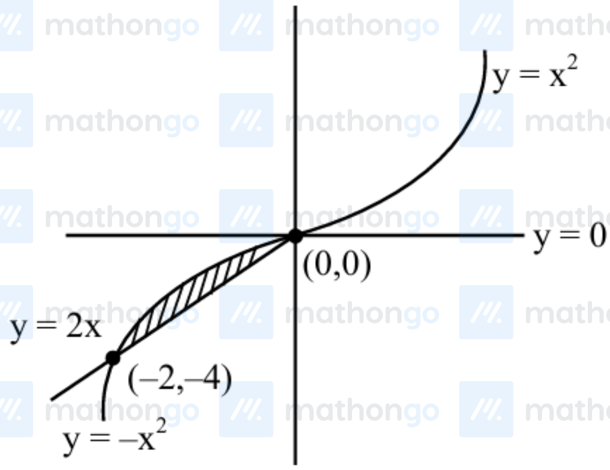
$$\int_0^6 \left(\left(7x - x^2\right) - \left(x^2 - 5x\right)\right) dx$$

$$\int_0^6 \left(12x - 2x^2\right) dx = \left[12\frac{x^2}{2} - \frac{2x^3}{3}\right]_0^6$$

$$\Rightarrow 6(6)^2 - \frac{2}{3}(6)^3$$

$$= 216 - 144 = 72 \text{ unit }^2$$

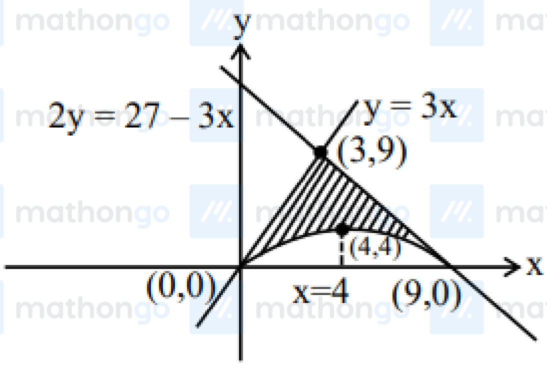
Q4



$A = \int_{-2}^{0} -x^2 - 2x = \frac{4}{3}$

Q5

$y = 3x, 2y = 27 - 3x \& y = 3x - x\sqrt{x}$



$$A=\int_0^3 3x-(3x-x\sqrt{x})dx+\int_3^9\left(\frac{27-3x}{2}-(3x-x\sqrt{x})\right)dx$$

$$A=\int_0^3 x^{3/2}dx+\int_3^9\frac{27}{2}-\frac{9x}{2}+x^{3/2}dx$$

$$A=\left[\frac{2x^{5/2}}{5}\right]_0^3+\frac{27}{2}[x]_3^9-\frac{9}{2}\left[\frac{x^2}{2}\right]_3^9+\left[\frac{2x^{5/2}}{5}\right]_3^9$$

$$A=\frac{2}{5}\left(3^{5/2}\right)+\frac{27}{2}(6)-\frac{9}{4}(72)+\frac{2}{5}\left(9^{5/2}-3^{5/2}\right)$$

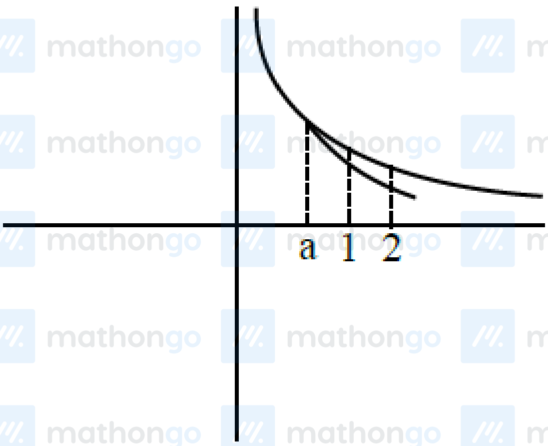
$$A=\frac{2}{5}\left(3^{5/2}\right)+81-162+\frac{2}{5}\times 3^5-\frac{2}{5}\times 3^{5/2}$$

$$A=\frac{486}{5}-81=\frac{81}{5}$$

$10\ A=162$

Ans. $=(2)$

Q6



area $\int_1^2\left(\frac{1}{x}-\frac{a}{x^2}\right)dx$

$$\left[\ell nx+\frac{a}{x}\right]_1^2$$

$$\ell\text{nn}\,2+\frac{a}{2}-a=\log_e 2-\frac{1}{7}$$

$$\frac{-a}{2}=-\frac{1}{7}$$

$$a=\frac{2}{7}$$

$7a=2$

$7a-3=-1$

Q7

$$\int_0^a f(x)dx = e^{-a} + 4a^2 + a - 1$$

$f(a) = -e^{-a} + 8a + 1$

$f(x) = -e^{-x} + 8x + 1$

Now $y = C_1 \mathrm{f(x)} + \mathrm{C_2}$

$\frac{\mathrm{dy}}{\mathrm{dx}} = \mathrm{C_1 f'(x) = C_1 (e^{-x} + 8)}$ ...(1)

$\frac{d^2y}{dx^2} = -C_1 e^{-x} \Rightarrow -e^x \frac{d^2y}{dx^2}$

Put in equation (1)

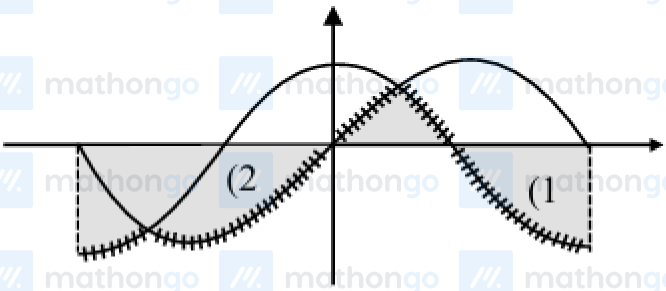
$$\frac{dy}{dx} = -e^x \frac{d^2y}{dx^2}(e^{-x} + 8)$$

$$(8e^x + 1)\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$$

Q8

$y = \min\{\sin x, \cos x\}$

$x -$ axis $\quad x - \pi \quad x = \pi$



$$\int_0^{\pi/4} \sin x = (\cos x)^0_{\pi/4} = 1 - \frac{1}{\sqrt{2}}$$

$$\int_{-\pi}^{-3\pi/4} (\sin x - \cos x) = (-\cos x - \sin x)^{-3\pi/4}_{-\pi}$$

$= (\cos x + \sin x)^{-\pi}_{-3\pi/4}$

$= (-1 + 0) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)$

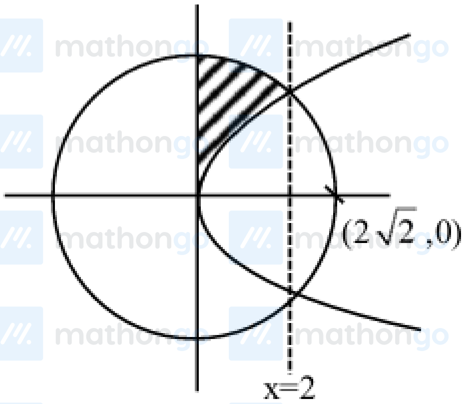
$= -1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}$

$$\int_{\pi/4}^{\pi/2} \cos x dx = (\sin x)^{\pi/2}_{\pi/4} = 1 - \frac{1}{\sqrt{2}}$$

$A = 4$

$A^2 = 16$

Q9



Required area $=\operatorname{Ar}($ circle from 0 to $2)-$ ar(para from 0 to 2)

$$=\int_0^2 \sqrt{8-x^2} d x-\int_0^2 \sqrt{2 x} d x$$

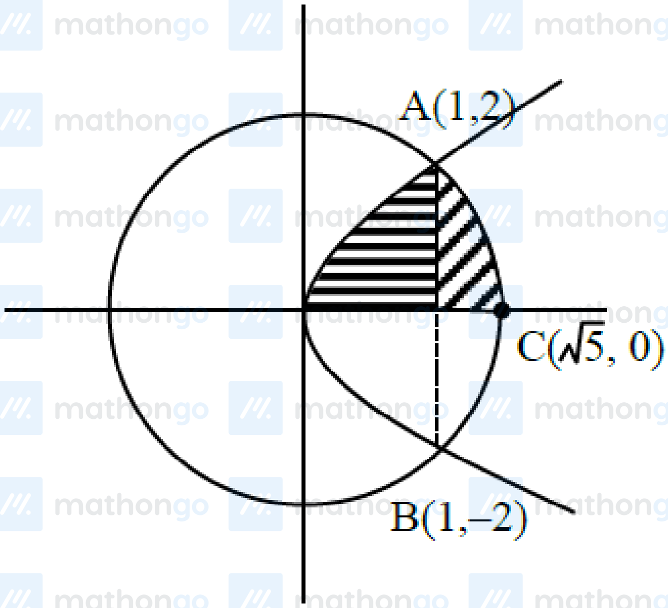
$$=\left[\frac{x}{2} \sqrt{8-x^2}+\frac{8}{2} \sin ^{-1} \frac{x}{2 \sqrt{2}}\right]_0^2-\sqrt{2}\left[\frac{x \sqrt{x}}{3 / 2}\right]_0^2$$

$$=\frac{2}{2} \sqrt{8-4}+\frac{8}{2} \sin ^{-1} \frac{2}{2 \sqrt{2}}-\frac{2 \sqrt{2}}{3}(2 \sqrt{2}-0)$$

$$\Rightarrow 2+4 \cdot \frac{\pi}{4}-\frac{8}{3}=\pi-\frac{2}{3}$$

Q10

$y^2=4 x$
$x^2+y^2=5$
$\therefore$ Area of shaded region as shown in the figure will be



$$A_1 = \int_0^1 \sqrt{4x}dx + \int_1^{\sqrt{5}} \sqrt{5-x^2}dx$$

$$= \frac{4}{3} \cdot \left[x^{\frac{3}{2}}\right]_0^1 + \left[\frac{x}{2}\sqrt{5-x^2} + \frac{5}{2}\sin^{-1}\frac{x}{\sqrt{5}}\right]_1^{\sqrt{5}}$$

$$= \frac{1}{3} + \frac{5\pi}{4} - \frac{5}{2}\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$\therefore$ Required Area $= 2\ A_1$

$$= \frac{2}{3} + \frac{5\pi}{2} - 5\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

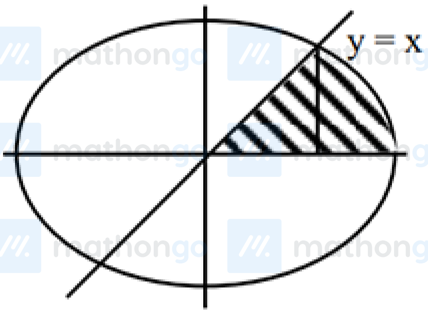
$$= \frac{2}{3} + 5\left(\frac{\pi}{2} - \sin^{-1}\frac{1}{\sqrt{5}}\right)$$

$$= \frac{2}{3} + 5\cos^{-1}\frac{1}{\sqrt{5}}$$

$$= \frac{2}{3} + 5\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$$

Q11

$\frac{x^2}{18}+\frac{y^2}{6}=1$



$$\frac{x^2}{18}+\frac{3x^2}{18}=1 \Rightarrow 4x^2=18 \Rightarrow x^2=\frac{9}{2}$$

$$\int_{\frac{3}{\sqrt{2}}}^{3\sqrt{2}} \frac{\sqrt{18-x^2}}{\sqrt{3}} dx$$

$$=\frac{1}{\sqrt{3}}\left(\frac{x\sqrt{18-x^2}}{2}+\frac{18}{2}\sin^{-1}\frac{x}{3\sqrt{2}}\right)_{\frac{3}{\sqrt{2}}}^{3\sqrt{2}}$$

$$=\frac{1}{\sqrt{3}}\left(9\times\frac{\pi}{2}-\frac{3}{2\sqrt{2}}\times\frac{3\sqrt{3}}{\sqrt{2}}-9\times\frac{\pi}{6}\right)$$

Required Area

$$=\frac{1}{2}\times\frac{9}{2}+\left(\frac{18\pi}{6}-\frac{9\sqrt{3}}{4}\right)\frac{1}{\sqrt{3}}$$

$$=\sqrt{3}\pi$$